we take two functions, *F* and *G*, and consider the differential of their product *FG*. We can write

$$d(FG) = FdG + GdF$$

or

$$d(FG) - GdF = FdG.$$

Taking the definite integral gives us

$$\int_a^b d(FG) - \int_a^b GdF = \int_a^b FdG$$

or

$$FG\Big|_a^b - \int_a^b GdF = \int_a^b FdG.$$

This is the standard formula that you may remember from calculus. But in quantum mechanics the limits of integration tend to span the entire axis, and our wave functions must go to zero at infinity to be properly normalized. Therefore, the first term of this expression will always evaluate to zero. With that in mind, we can use a simplified version of integration by parts:

$$\int_{-\infty}^{\infty} F\frac{dG}{dx}dx \;=\; -\int_{-\infty}^{\infty} \frac{dF}{dx}Gdx.$$

This form is correct as long as *F* and *G* go to zero appropriately at infinity, so that the boundary term becomes zero. You will do yourself a big favor if